```
-/+ buffers/cache:   435      1286
Swap:          3967            0      3967
```

The *df* command gives the amount of file space used, and the free space in the currently mounted file systems. The *-h* (human readable format) option will give the size in MB or GB.

```
[root@sys70 ~]# df -h
Filesystem        Size    Used   Avail   Use% Mounted on
/dev/mapper/VolGroup00-LogVol00
                  222G     25G    186G    12%  /
/dev/hda1         99M      36M     65M     32% /boot
tmpfs             862M      0     862M     0%  /dev/shm
none              861M     40K    861M     1%  /var/lib/
xenstored
```

The time elapsed since the last reboot can be seen with the *uptime* command. It also shows the number of users currently logged in, and the load averages (the average number of processes active during the last one, five and 15 minutes).

```
[root@sys70 ~]# uptime
 12:08:36  up   3:36,   user,  load average: 1.27, 0.82, 0.40
```

The last command can be used to display the list of users last logged-in on your system, including the date, time and the terminal from which they logged in. The command with a user name as parameter will display the particular user's log-ins.

```
[root@sys70 ~]# last sn
sn   pts/6   :0.0  Tue Jan 22 15:09 - 15:09  (00:00)
sn   :0            Tue Jan 22 15:09 - down   (01:46)
sn   :0            Tue Jan 22 15:09 - 15:09  (00:00)
sn    pts/3  :0.0  Mon Jan 21 15:33 - down   (01:22)
...................
```

The *lastlog* command shows all the users and their last login times. The system reboot history can be shown by *last reboot*. The *who* command displays the names of *users* logged in to the system currently, the time they logged in, and their terminals:

```
[root@sys70 ~]# who
root           pts/2          2013-01-28 10:35 (:1001.0)
```

You can display only the names of all the users currently logged in, with the users command. You can find out who all are currently logged into the system and what they are doing by using the command *w;* the output is similar to a combination of the *who, uptime* and *ps –a* commands. The output given by *w* is only an approximation; it does not give an entirely correct picture. You can use *w* to give details about a particular set of users also.

The *lspci* utility can be used for displaying information about the PCI buses and the devices connected to them. It gives data about any user who has logged in through the console and not through a (GUI) terminal window.

The *dmesg* program can help users print out their boot-up messages, and will also help in viewing all the hardware devices detected by the system.

The *proc* file system

The Linux *proc* virtual file system is not associated with a hardware device, and can be used to get detailed information about a system's hardware, the system status, and also to change the kernel parameters. The */proc* directory contains all the information about the currently running Linux system. There is a sub-directory for each running process, and the sub-directories are named after the process' ID (PID) number. Each of the sub-directories has many standard files, with each giving a set of information. Since */proc* is a virtual file system, the files in it are not actual files on a physical device, and are zero bytes in size. The contents are generated by the kernel when the file is read by the user. You can view specific files containing hardware information in the */proc* directory to get the hardware and the status of the running system.

The */proc/version* file displays the version, and is more accurate and complete than *uname*:

```
[root@sys70 ~]# cat /proc/version
Linux version 2.6.18-348.1.1.el5xen (mockbuild@builder17.
centos.org) (gcc version 4.1.2 20080704 (Red Hat 4.1.2-54))
#1 SMP Tue Jan 22 17:49:28 EST 2013
```

Also, */proc/cpuinfo* gives detailed information about the CPU, including the family, clock speed, number of cores, cache memory, etc:

```
[root@sys70 ~]# cat /proc/cpuinfo
processor      : 0
vendor_id      : GenuineIntel
cpu family     : 6
model          : 23
model name     : Intel(R) Core(TM)2 Duo CPU     E7500  @
2.93GHz
stepping       : 10
cpu MHz        : 1596.000
cache size     : 3072 KB
...................
```

You can view the list of interrupts as follows:

```
[root@sys70 ~]# cat /proc/interrupts
          CPU0        CPU1
  1:      24           0       Phys-irq  i8042
  8:      1            0       Phys-irq  rtc
  9:      0            0       Phys-irq  acpi
 12:     598           0       Phys-irq  i8042
```